

Motion2Scene: Execution-Conditioned Teaching for Perceptive Humanoid Traversal

Working method revision

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Abstract

A motion repertoire is useful for traversal only when the robot can execute a suitable alternative and select it from information available before commitment. We develop Motion2Scene, an execution-aware curriculum for humanoid overhead traversal with a frozen low-level tracker. Executed body envelopes construct solution-preserving geometric proposals; complete physical schedule branches supply passage-first, measured-time teaching targets. A 114-dimensional causal floor/ceiling-history policy can wait at three legal phases before one adaptation and recovery. Historical replay and information-consistent teaching are evaluated as separate development extensions. In 42 forced-schedule simulation episodes on six development contexts, seven qualified schedules provide a passing alternative for every context; the strongest constant passes five. A prior-derived option and an authored sustained option solve complementary constraints. A 36-episode policy comparison on the same six development contexts shows that adding the complementary teaching data raises learned passage from five to six contexts, matching a script retuned on the same data. Both updated policies take 0.50 s longer on each of two previously solved passages. These results establish a constructive option-and-data extension with a measured efficiency tradeoff; independent curriculum benefit and held-out improvement remain open.

1 Introduction

A humanoid may track diverse motions yet fail to use them in a constrained room. An overhead passage requires a whole-body transition early enough to clear its front edge, a sufficiently low executed shape throughout its length, and a legal return to locomotion. A reference pose fitting beneath the obstacle establishes none of these conditions. A training scene also provides little decision value when its relevant sensor evidence arrives after the useful transition.

The teaching problem is to expose decisions that matter. An obstacle cleared by every motion may teach timing preferences but not the need to change body shape; an obstacle defeating every motion cannot be solved by improving selection alone. Our approach constructs constraints from the swept body geometry of executed alternatives. A useful contrast leaves one schedule geometrically clear while intersecting another, concentrating physical labeling on candidate decisions where the available responses differ.

We retain the frozen SONIC tracker [11] and expose qualified entry-reference-return schedules built from generated and authored motions. The selector learns when to commit from egocentric floor and ceiling history. Crucially, WAIT retains later choices: its target must evaluate a complete continuation, and that continuation must be selectable from the information available at the later decision. Figure 1 shows why selecting between alternatives matters: the prior-derived and sustained responses solve different constraints.

Motion-conditioned scene construction has precedent in Learning from Hallucination (LfH), Learning from Learned Hallucination (LfLH), and LfH-CP [1, 2, 3]; PAIRED and ACCEL guide environment design through performance gaps [4, 5], while Prioritized Level Replay selects previous environments according to estimated learning potential [6]. Motion2Scene links construction and learning through measured humanoid execution. Executed contrasts propose scenes, complete physical continuations teach commitment, and pre-update student outcomes provide a separate replay signal. The geometric candidate queues remain fixed throughout acquisition.

Our technical contributions are execution-conditioned contrast construction and complete-continuation supervision for multi-phase perceptive commitment. We also formulate an observation-consistent teacher on the finite decision tree and compare replay signals within a common learner. Experiments distinguish the effects of extending physical capability, acquiring teaching encounters and reweighting those encounters. The completed

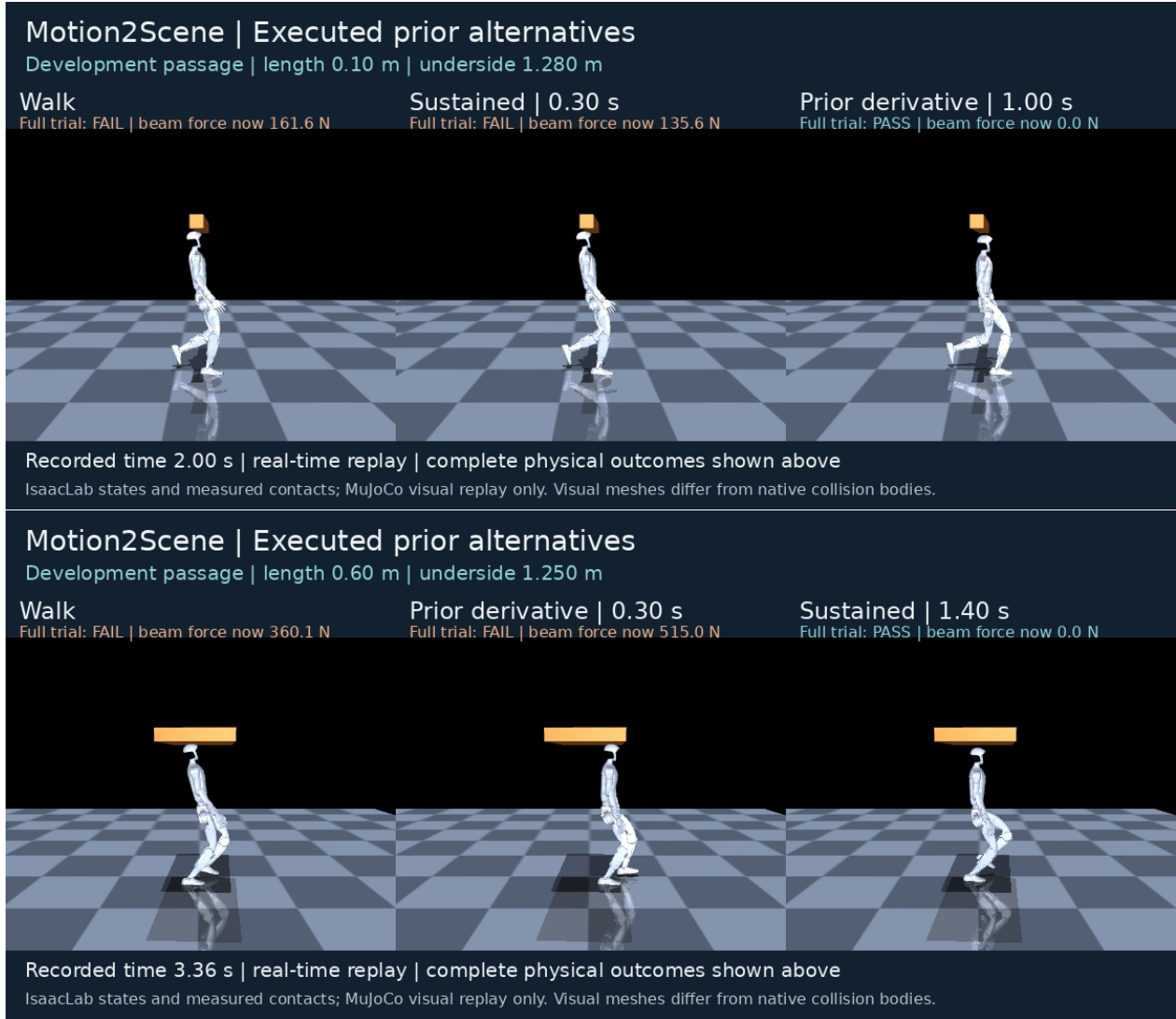


Figure 1: Complementary development capability. Top: the prior-derived schedule passes an early beam that walking and sustained adaptation contact. Bottom: sustained adaptation passes a different passage that the prior derivative contacts. Images replay recorded Isaac Lab states and measured native forces; MuJoCo provides visualization only. Complete episode measurements determine the labels.

development results establish complementary responses and a closed-loop teaching-data benefit with a time regression. The broader acquisition study tests whether construction and supervision produce a repeatable downstream advantage.

2 Related Work

Inverse construction from motion. LfH constructs obstacle environments around motion plans; LfLH learns the inverse map through a planner decoder; LfH-CP uses critical spatial and temporal constraints to obtain varied dynamic navigation examples [1, 2, 3]. LfH already collects executed motions in obstacle-free space. Motion2Scene retains this inverse-construction idea and tests a more specific teaching design: contrasting complete humanoid schedules through executed body geometry, then physically labeling their transitions and recovery in the proposed scene. Complete continuation targets connect these measurements to legal perceptive decisions. The benefit over simpler construction remains an empirical question.

Complete schedule	Entry phase (s)	Return phase (s)
Neutral	—	—
Short, early / late	0.30 / 1.40	5.30
Sustained, early / late	0.30 / 1.40	5.10
Projected/spliced prior, early / late	0.30 / 1.00	5.30

Table 1: Seven physically qualified schedules from four references. Paired entries denote two separately qualified schedules. One adaptation per episode; mandatory recovery is part of the option.

Curricula and imitation. PAIRED optimizes a protagonist–antagonist gap and ACCEL evolves environments through regret-based editing [4, 5]. Prioritized Level Replay (PLR) selects environments using estimates of future learning potential, including temporal-difference error [6]. Our replay component reweights fixed acquired encounters, gated by measured sensor delivery and an actual matched student failure or time gap, without editing their geometry. DAgger motivates labeling student-visited states [12], but a one-entry schedule bank may already cover every neutral decision prefix; adding a student episode does not by itself establish new state coverage. Our refreshed-error control uses supervised regret residuals on recorded teacher inputs; it adapts the learning-potential idea to supervised fitting rather than reproducing PLR’s RL algorithm.

Teaching under asymmetric information. A privileged expert can demand decisions that a partially observed student cannot reproduce. Adaptive asymmetric DAgger jointly modifies the expert and trainee [7]; Student-Informed Teacher Training penalizes teacher–student action differences and adds an alignment step [8]. Guided Policy Optimization likewise co-trains an aligned privileged guider and learner [9]. Our finite continuation extension addresses this established information mismatch through shared actions within declared observation groups, using complete measured schedule tables rather than jointly trained RL policies.

Perceptive humanoid control. SONIC supplies motion tracking and contextual kinematic planning [11]. Kimodo already demonstrates generated G1 motions tracked by GEAR-SONIC in simulation [10]. Our research question concerns how executed alternatives construct useful teaching scenes and inform perceptive commitment decisions. Perceptive Humanoid Parkour combines motion composition, tracking experts and a perceptive student [13]; HumanoidPF combines obstacle representations with realistic and procedural scenes [14]. Our comparison concerns training-data construction for a common frozen tracker, option set and sensor interface. It does not reproduce their broader navigation or parkour scope.

3 Problem and Executable Interface

We separate two questions: does the option bank contain a passing response, and can the sensor policy select that response with low measured passage cost? A scene q contains an overhead constraint or two ordered constraints along a fixed approach. From robot state x_t , executed command history h_t , and simulator seed ξ , the frozen tracker K produces

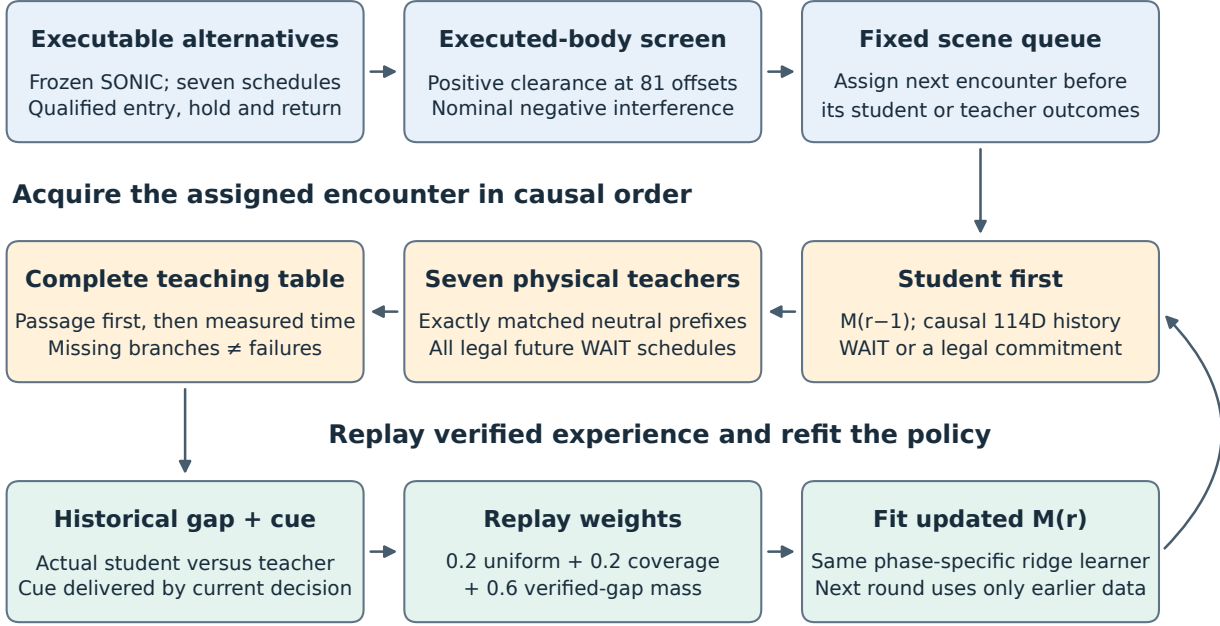
$$\tau_s = \text{Execute}(K, x_t, h_t, s, q, \xi), \quad (1)$$

where s is a complete qualified schedule. The reference bank contains neutral, short and sustained authored crouches, and one projected/spliced prior derivative. Table 1 defines the seven available schedules. Neutral and the longer carrier were generated with the installed Kimodo prior. Short and sustained are local authored variants of that neutral, not independently generated skills. The derivative retains the provenance of its separate constrained prior sample and its explicit repair operator.

The high-level policy decides while neutral at reference ticks $k \in \{15, 50, 70\}$ (0.30, 1.00 and 1.40 s). It may commit to a currently legal schedule or WAIT. After commitment, the qualified schedule controls holding and return; no second entry, adaptation-to-adaptation switch, arbitrary duration, stop or steering command is available. Different request phases can precede the same physical adaptation onset, so entry time alone is not a measured capability improvement.

A six-second, 180-frame 30 Hz reference loads as 299 native 50 Hz samples. Each complete bounded episode records 298 synchronized control rows and 1,192 physics substeps. The last sensor capture is at elapsed 5.94 s, before the wrap callback. At a reference tick k , the preaction sensor packet is captured at $(k - 1)/50$ s. Entry/return phases use the reference clock; passage time uses the physical recording clock.

Construct from execution



No positive gap: 0.8 uniform + 0.2 coverage. Geometry remains fixed during the replay comparison.

Figure 2: Execution-conditioned teaching. Executed alternatives screen candidate scenes at 81 geometric offsets. Each acquired scene supplies a pre-update student execution and seven matched teacher branches. Complete continuations supervise commitment timing; uniform, coverage and observable historical-gap components weight the next fit.

Passage requires initial approach upstream of every enabled beam, ordered full body-origin crossing and the registered 0.30 s stabilization hold, legal recovery, no fall/reset, and no disallowed environmental contact throughout the recorded horizon. Explicit body/counterpart forces are measured at 200 Hz; the registered contact limit is 1 N, with permitted foot-floor support and internal self-contact reported separately. A crossed finish with insufficient remaining hold is a finite-horizon task failure. An admitted early contact, fall or refusal can establish failure even when later supervision is unavailable; unverified outcomes remain unknown. Mechanical work is unavailable in this interface because applied actuator torque is not measured; it is stored as null. Teaching cost uses measured successful passage time, without an invented crouching or energy penalty. Specifically, passage time ends at crossing plus stabilization; success additionally requires recovery and acceptable contact through the remaining horizon. Thus a 2.72 s passage may precede a 5.30 s return phase. Passage time is not the duration of the complete adaptation-recovery maneuver, and does not measure readiness to begin a second adaptation.

For each fixed task, $\max_s \mathbf{1}[\tau_s \text{ passes}]$ measures the capability of this finite bank. Actual policy completion additionally depends on choosing a usable schedule. A teacher search over all seven complete schedules establishes the bank's tested capability, not a globally optimal humanoid response.

4 Execution-Conditioned Teaching

Figure 2 connects the construction, physical teaching and replay stages. The same qualified schedules and observation interface serve all acquisition arms.

4.1 Qualified executions construct candidate scenes

Qualification verifies each complete schedule from the relevant neutral approach, including entry, maintained adaptation and return. Its measured native body poses transform conservative outer primitives and inner primitive

witnesses. We construct scenes using these executed envelopes, not reference heights. For a positive schedule s^+ and a negative alternative s^- , the analytic screen is

$$\min_{\delta \in \mathcal{U}} c_{s^+}^{\text{outer}}(q \oplus \delta) \geq m, \quad c_s^{\text{inner}}(q) \leq -m. \quad (2)$$

The six-second development screen uses $m = 0.01$ m and 81 combinations of world x/y offsets $\{-0.02, 0, 0.02\}$ m, underside offsets $\{-0.005, 0, 0.005\}$ m, and yaw offsets $\{-0.02, 0, 0.02\}$ rad. Positive clearance is required across this finite set; negative interference is required only at nominal placement. These are sampled geometric proposal conditions, not physical robustness certificates. Obstacles can change the subsequently executed trajectory.

The primary candidate pool is drawn before acquisition outcomes. Target-only assigns its positive schedule by a frozen random draw before geometric queries. Uniform uses the first robust positive in the original draw order, while analytic contrast searches distinct positive/negative pairs and maximizes minimum slack. These queues differ in geometric eligibility; target-only does not use a negative outcome. Common-support quotas select four distinct available scene strata per corpus in frozen queue order; no physical failure causes replacement or geometry re-selection. The analytic and observation curriculum arms use exactly the same candidate IDs, order and teacher schedules, so that their comparison isolates replay under the common acquisition design.

The reference-envelope comparator applies native URDF forward kinematics to the commanded full schedules, including entry and return, and uses identical collision shapes, candidate scenes, margins and placement offsets. We first check this kinematic map against native body poses using recorded root and joint states. The comparison then changes only commanded versus executed body poses; both constructors receive the same complete physical labeling and downstream learner.

4.2 Causal observation and complete-continuation teaching

The student receives 114 named values: 100 sensor/history, phase and robot-state features, seven preaction active-schedule indicators and seven legality entries. Sixty-five ideal mounted PhysX rays and normals accumulate for at most two seconds or 101 frames in an egocentric floor/ceiling representation with separate observed/unknown masks. Scene parameters, object identities, physical labels and postdecision actions are excluded. This is an ideal simulated range interface, not a noisy depth-camera result or dense vertical-free-space guarantee.

Each scene is executed under all seven forced schedules with a matched physics seed. Before a teacher branch can label a decision, its recorded state, prior actions, reference tokens, causal sensor history, feature vector and legality must match the forced-neutral branch’s preaction prefix exactly. This supplies bootstrap targets even without an actual student episode. Matching an actual student’s visited neutral prefix is an additional requirement for gap-based replay below. For action a at phase k , let $\mathcal{S}_k(a)$ be all registered complete schedules sharing that immediate decision. A current commitment has its corresponding schedule; WAIT includes neutral continuation and every legal later-entry schedule. A complete WAIT target therefore cannot be replaced by the forced-walking outcome.

Every expected continuation must have an admitted outcome before its immediate action table is complete. From a complete table with a successful continuation,

$$s_k^* = \arg \min_{s \in \bigcup_a \mathcal{S}_k(a), \tau_s \text{ passes}} T_s. \quad (3)$$

The immediate teacher action is the first action of s_k^* . An action’s value uses its best passing complete continuation, or its measured failure if none passes. Missing branches and unavailable later phases are not failures. Tables without a successful continuation provide no fitting target. Ordinarily, fitting also requires consequential passage or cost differences. A narrow measured-tie initialization rule below handles an otherwise unsupported phase; all physical recordings and costs remain in the data.

The common student fits a separate linear ridge head at each legal phase to negative measured regret. For a complete decision table, failure has regret one; a passing action has

$$R_k(a) = \frac{T_k(a) - T_k^*}{\max_{b \in \mathcal{A}_k, b \text{ passes}} T_k(b)}. \quad (4)$$

Per-phase input normalization uses a standard-deviation floor of 0.05; intercepts are unpenalized. Regression weights are normalized per legal action. Runtime scores are negative predicted regrets: masked score argmax selects the legal action with minimum predicted regret. These unconstrained estimates are not success probabilities, and regression does not preserve the teacher’s lexicographic success preference as a guarantee. The common penalty $\lambda = 10$ was selected on five development layouts from a registered finite sweep, before primary acquisition.

At bootstrap, a phase with only complete, equally successful and equal-time alternatives receives zero regret targets and deterministic tie breaking. Once consequential examples arrive, they determine that phase’s fit. The exact rule and its development history appear in Appendix B.

4.3 Observation-consistent continuation supervision

Scene-wise schedule search can overvalue WAIT when later successful actions depend on information unavailable to the student. We implement a finite information-group teacher to test and address this failure. Let I_k contain encounters with exactly equal available 114-dimensional student vectors at phase k . For this dynamic program, successive partitions must refine: later groups must retain earlier distinctions. Inputs that violate this condition require an explicit memory representation or a different policy solver.

Starting at the last phase, retain every legal commitment that passes in all members of I_k . Retain WAIT only when every successor group admits a common passing continuation. In notation,

$$\mathcal{A}_k^+(I_k) = \{a \in \mathcal{A}_k^{\text{commit}} : Y_i(a) = 1 \ \forall i \in I_k\} \cup \{\text{WAIT} : \mathcal{A}_{k+1}^+(J) \neq \emptyset \ \forall J \subseteq I_k\}, \quad (5)$$

where J ranges over successor information groups; at the last phase WAIT means the complete neutral schedule. $Y_i(a)$ is the measured full-schedule outcome. Incomplete continuation tables withhold supervision. Among passing policies, choose the one minimizing mean measured passage time within the group; retain all passing actions and all exact time minimizers as set-valued targets. This enforces equal actions for equivalent information while allowing later observations to distinguish tasks.

When no common policy passes the entire group, the teacher maximizes attainable passage count N_a before minimizing the successful-time sum C_a . A commitment uses its measured schedule; WAIT sums the optimal successor-group values. Backward induction retains all equal optima under uniform empirical encounter mass. Let $N^* = \max_a N_a$ and let C_*, C^* be the minimum and maximum costs among actions attaining N^* . For $N^* > 0$, shared regret targets are

$$R_I(a) = \begin{cases} (N^* - N_a)/N^*, & N_a < N^*, \\ (C_a - C_*)/(N^* C^*), & N_a = N^*. \end{cases} \quad (6)$$

Positive passage times make every equal-count time regret smaller than losing one passage. Singleton groups recover the original regret exactly. These explicit targets train the same ridge equations; unknown or all-failing groups provide no passing supervision. The teacher is a development extension; the ongoing acquisition comparison retains its original scene-wise teacher.

4.4 Historical physical gaps prioritize replay

At acquisition round r , the student model M_{r-1} is committed with exactly bootstrap and earlier-round teacher data before the current student’s scene is executed. All seven current teacher branches follow that actual student episode. This order prevents current-scene labels from entering the model used to measure its gap. Every arm uses the same student-before-teachers encounter structure and branch budget.

For a matched neutral phase with a complete feasible teacher, compare the actual full student outcome with the teacher’s best continuation:

$$g_{r,k} = \begin{cases} 1, & \text{verified student failure,} \\ \max\left(0, \frac{T_{M_{r-1}} - T_k^*}{\max(T_{M_{r-1}}, T_k^*)}\right), & \text{student passes.} \end{cases} \quad (7)$$

Unverified, unmatched or unavailable outcomes receive no gap priority. Recorded ray endpoints are associated with beam geometry only for training-time cue scoring. The primary rule requires each enabled beam’s surface cue to have been delivered by the current decision’s physical timestamp. It does not require an underside return and does not prove that two tasks are distinguishable. A future-entry cue is separately recorded and is never credited as current information. Safe WAIT without a current cue remains eligible for ordinary teacher training, but receives no regret priority under this rule.

Replay mixes 0.2 uniform, 0.2 declared option/length/phase/entry/return coverage, and 0.6 verified-gap mass. We normalize encounter mass before distributing it across supervised phases, preventing extra WAIT opportunities from tripling an encounter’s weight. An encounter’s gap is the maximum of its eligible phase gaps. When all gaps are zero or unavailable, only the 0.6 gap component falls back to uniform: the resulting mixture is *0.8 uniform plus 0.2 coverage*. Missing supervision gets zero fitting weight, not a fabricated label.

Seed	Executed contrasts	Reference contrasts	Selected pairs retained
93201	77	206	59/206
93202	89	241	67/241
93203	65	198	43/198

Table 2: Common-pool geometric construction comparison. Retention requires the reference-selected positive to clear all 81 offsets and its selected negative to intersect nominally under executed envelopes, using the same 1 cm margins. These are geometric pair counts, not physical passage rates.

Coverage uses a joint key consisting of the best continuation’s reference, ordered enabled-beam length bins, current phase, entry tick and return tick. Length edges are $\{0.25, 0.5, 1, 2\}$ m; an exact edge belongs to the upper bin. Let n_b count encounters containing key b , counted once per encounter, and $v_{ik} = 1/n_{b(i,k)}$ for its supervised phases K_i . Define

$$u_i = \frac{1}{|K_i|} \sum_{k \in K_i} v_{ik}, \quad p_{ik}^{\text{cov}} = \frac{u_i}{\sum_j u_j} \frac{v_{ik}}{\sum_{\ell \in K_i} v_{i\ell}}. \quad (8)$$

The dimensions form one joint bin rather than separately averaged marginal distributions. The coverage term contributes $0.2p_{ik}^{\text{cov}}$.

A historical gap retains its generating model and round age. Our development controls compare uniform mass, coverage alone, their 0.8/0.2 mixture, and refreshed supervised-error priority. The latter replaces gap priority by the mean squared residual between predicted and measured regret across legal actions, recomputed after each fit. It uses the same encounter/phase normalization and fallback as historical replay. This is a cheap fitting signal; physical gaps continue to require actual executions of their generating models.

5 Experiments

All completed experiments below use development data: six fixed traversal contexts, independently acquired teaching encounters, and separate motion-option and perception studies. The six fixed contexts test empty, short, sustained, early-commitment, two-beam and complementary passages. Physical schedule tables, new closed-loop policy executions and recorded-input readouts are reported separately. The common acquisition learner is the phase-specific ridge model with penalty 10; the tracker and seven-schedule bank remain fixed. Larger acquisition checkpoints and reserved evaluation are described at the end of this section. Appendix E documents the reusable recordings.

5.1 Does execution change the constructed teaching constraints?

We extend the shared geometric pool to 3,840 candidates: 256 draws per stratum, five strata and three seeds. Native forward kinematics reproduces recorded body positions within 2 micrometres, supporting a controlled reference-versus-executed envelope comparison. Table 2 shows that only 169 of 645 reference-selected positive/negative pairs retain the same geometric contrast under measured execution. The chosen positive retains robust clearance in 182 cases; the chosen negative retains nominal intersection in 624. Thus the main discrepancy is whether the proposed passing response retains its clearance. Nevertheless, 207 reference-selected scenes have some executed contrast, possibly using a different pair. Complete physical teacher branches remain necessary to measure useful-task yield and downstream passage.

Does construction acquire tasks that require adaptation? At one acquired encounter per corpus, all twelve original corpora complete their pre-update student and seven teacher executions. Across three seeds, feasibility-screened uniform acquires three tasks on which every schedule passes. Target-only, executed contrast and observation curriculum each acquire three tasks requiring adaptation: the neutral branch fails while a supported adaptation passes. The pre-update student passes all three uniform tasks and fails all three tasks of each other constructor. These are constructor-specific acquisition difficulties, not held-out policy comparisons. All 96 encounter episodes have known outcomes, excluding bootstrap. Target-only matches executed contrast at this first budget, so the result establishes adaptation-requiring teaching examples without yet isolating a contrast-targeting benefit.

Constructor	M1	M2	M4
Feasibility-screened uniform	0/3	0/6	2/12
Target-only	3/3	3/6	4/12
Executed contrast	3/3	6/6	12/12
Executed contrast with replay	3/3	6/6	12/12
Measured physics steps per arm	53,640	82,248	139,464

Table 3: Adaptation-required acquired tasks across three corpus seeds per arm. Neutral must fail and at least one adaptation pass; all assigned tasks are bank-solvable. Costs include each corpus’s bootstrap, all teacher branches and pre-update student executions. Checkpoints share nested acquisition prefixes; their costs are not additive. Executed-contrast and replay arms share candidate identities and order, with no measured teacher-outcome disagreements. These task yields are distinct from common-set policy performance.

At the completed M2 checkpoint, executed contrast and observation replay each retain six adaptation-required tasks out of six acquired across three corpora; target-only retains three of six and screened uniform zero of six. Every acquired task is bank-solvable. Each arm consumes 57,216 measured encounter steps, plus 25,032 steps for its three independent bootstraps: 82,248 steps per arm and 27,416 per corpus. All 192 encounter outcomes are known. The difference now supports increased physical adaptation-required task yield relative to target-only at this budget. It still does not establish useful perceptive selection: a fixed late sustained schedule passes every M2 acquired task across all four arms.

The completed M4 audit retains all 468 acquisition episodes, including 84 bootstrap teachers, 336 encounter teachers and 48 pre-update students: 557,856 measured physics steps, or 139,464 per arm. Executed contrast and replay each acquire 12/12 adaptation-required tasks, compared with 4/12 for target-only and 2/12 for screened uniform (Table 3). The contrast versus target-only differences are 3/4, 3/4 and 2/4 tasks in the three corpora. All assigned tasks remain bank-solvable; the complete prefixes retain 117 failed teacher branches and ten failed students, with no unknown branch outcomes. The reference arm is acquiring its own physical prefix, and common-set M8 policy evaluation remains pending.

Do acquired tasks require different responses? Every individual M4 corpus still admits one fixed schedule covering all four acquired tasks (Fig. 3). Pooled within each arm, one fixed schedule also covers all twelve tasks: either prior for uniform, executed contrast and replay, and late sustained for target-only. Across all four constructors, the bank covers 34/34 distinct geometry/seed conditions, whereas the best fixed schedule covers 32/34; one prior and one sustained schedule together cover the full pool. Thus complementary responses exist across the acquired data, but no individual training corpus yet requires choosing between them. Appendix A illustrates a same-seed pair with disjoint passing sets and different recorded upper-hit features at 0.30 s. This is physical capability and recorded observation evidence, not a demonstration of robust perceptual selection. The audit adds no physics or training.

Cross-corpus decision readout. We test the acquired M1 models on recorded teacher branches from the other acquisition seeds, using identical validation sets across training arms and removing duplicate geometry/seed pairs. Eight unique short-stratum scenes yield 5, 6 and 5 validation contexts for training seeds 93201–93203. Uniform-trained models select passing branches in 2/5, 5/6 and 4/5 contexts; target-only gives 4/5, 5/6 and 4/5; executed contrast and observation replay both give 5/5, 5/6 and 4/5. The contrast-trained models always choose a prior schedule and match the fixed prior’s counts. A fixed late sustained schedule covers all eight unique scenes. Thus this recorded-data comparison shows a one-seed construction effect, but neither a replay gain nor improved selection over the best fixed response within these tables. Folds reuse validation scenes; these are not new physical policy evaluations.

At M2, the cross-corpus pool contains 17 unique scenes: eight from the first acquisition round and nine from the second. The corresponding validation sets contain 11, 12 and 11 contexts. Uniform, executed contrast and observation replay all select passing branches in 11/11, 11/12 and 10/11 contexts; target-only gives 10/11, 11/12 and 10/11. The three tied arms choose a constant prior schedule within each corpus. A fixed late sustained schedule passes all 17 tasks. Thus the increased adaptation-required yield at M2 has not produced better selection than uniform or a strong fixed response. Because the validation pool grows between checkpoints, these readouts are not a fixed-test learning curve.

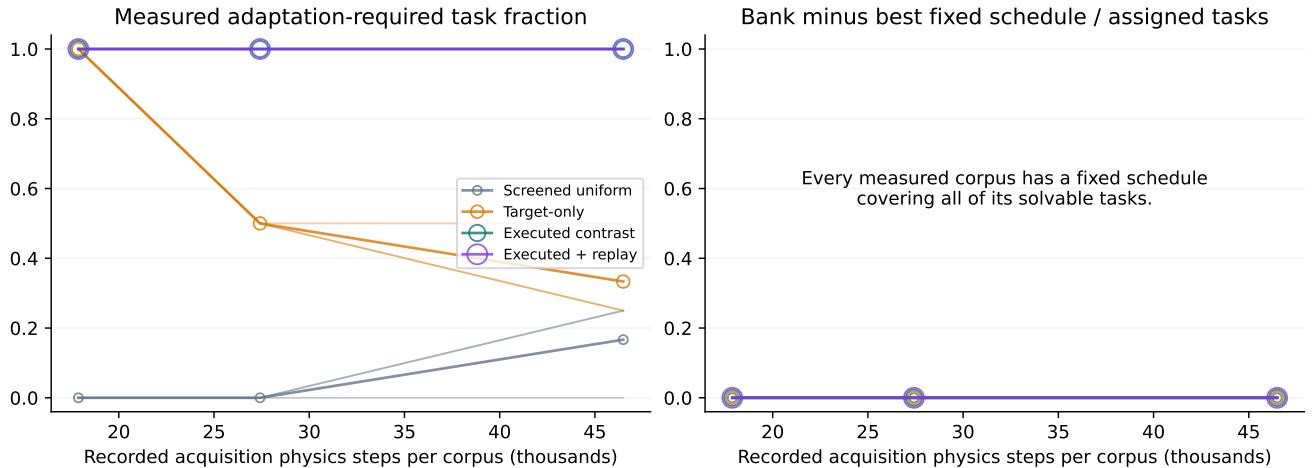


Figure 3: Completed M1/M2/M4 training-task measurements against actual acquisition steps per corpus, including bootstrap. Thin lines show individual corpora and bold lines their means; there are three corpora per arm. Increased adaptation-required yield (left) coexists with a fixed schedule covering every task within each corpus (right). Constructor-specific tasks are not a common policy evaluation set.

Schedule / entry phase	Empty	Short	Long	Early	Two beams	Complement
Neutral	5.36	fail	fail	fail	4.14	fail
Short, 0.30 / 1.40	5.86	4.04	fail	fail	4.20	fail
Sustained, 0.30 / 1.40	5.92	4.06	4.64	fail	4.52	4.20
Prior derivative, 0.30	5.36	3.62	4.48	2.74	4.12	fail
Prior derivative, 1.00	5.38	3.62	4.50	2.72	4.12	fail

Table 4: Successful passage times (seconds) for all 42 nominal forced-schedule development episodes, seed 8731. Paired authored entries are separately executed schedules with identical listed outcomes. “Fail” is a measured task failure. Finish planes differ across scenes; compare times within columns.

At M4, 34 distinct geometry/seed conditions give other-seed validation folds of 22, 23 and 23 contexts. Uniform, executed contrast and replay again tie, now at 21/22, 21/23 and 22/23 passing branch selections; target-only gives 16/22, 21/23 and 16/23. The three tied arms still choose a constant prior within each corpus and match the fixed prior’s passage counts. The folds reuse conditions and the pool grows across checkpoints. These saved-policy, recorded-branch readouts add no physical policy executions and provide no evidence of better selection than uniform or the fixed prior at M4.

5.2 Additional executable capability

Table 4 reports all seven schedules on five original contexts and one separately constructed complementary passage at seed 8731: 42 physical simulation episodes, 12,516 aligned control/sensor rows and 50,064 substeps. The prior-derived schedules pass the early beam, while all authored short and sustained schedules fail. On the complementary passage, both sustained schedules pass in 4.20 s with zero beam force, while both prior schedules fail torso contact with maxima of 514.955 and 400.533 N. All 42 measurements are complete; 28 pass and 14 fail. The finite option union covers six contexts, while the best constant covers five. This is a constructed development capability distinction.

The complementary beam was selected by a registered executed-geometry search: 0.60 m long, 1.25 m under-side, with sustained outer clearance at least 13.201 mm over the finite offsets. Both prior nominal inner witnesses interfere. All 95 selected primitive queries agree with independent uncapped full-capsule checks. Only nominal placement has physical evidence. The course column is deliberately retained: neutral passes it, so that course demonstrates two-beam instrumentation and continuation but does not require adaptation or prove multiple transitions.

The 18 neutral phase tables are complete, with 17 consequential training targets. In the complementary

Policy, seed 8732	Empty	Short	Long	Early	Two beams	Complement	Pass
Learned, six contexts	5.38	4.10	4.68	2.72	4.12	4.54	6/6
Script, six contexts	5.40	4.10	4.68	2.72	4.12	4.54	6/6
Learned, five contexts	5.38	3.60	4.18	2.72	4.12	fail	5/6
Script, five contexts	5.40	3.60	4.18	2.72	4.12	fail	5/6
Constant prior, 0.30 s	5.38	3.60	4.18	2.72	4.12	fail	5/6
Always walk	5.40	fail	fail	fail	4.16	fail	2/6

Table 5: All 36 actual closed-loop development assignments. Entries are successful passage times in seconds or measured task failures, including one known partial walking failure. Compare times within columns; all six layouts were previously inspected. The updated learner and script trade an added passage for slower choices on two earlier contexts.

scene, WAIT at 0.30 s and 1.00 s is backed by the actually passing sustained schedule at 1.40 s. Ray-only reconstruction reproduces all 114 features. Useful sensor features differ from the original contexts, but no single map cell jointly observes floor and ceiling inside the student’s corridor at these decision phases. Thus the finite example is observable in its recorded summary without demonstrating complete vertical free-space perception.

Repeatable option extraction. We additionally compare four fresh prior samples with four authored variants, sharing the neutral carrier and applying the same joint-limit projection and neutral transition splice to every candidate. The authored variants use four fixed crouch-duration windows; the prior samples use fixed route, end-point and low-pelvis conditioning with independent generation seeds. Both procedures request 0.085 m lowering through their respective operators. Each candidate receives two qualification executions, entering at 0.30 or 1.00 s and returning at 5.30 s. All executions load the same nine-reference bank and use one common physics seed. Both procedures qualify all four candidates and all eight assigned schedules under the unchanged physical criteria; the shared neutral also passes. This 17-episode test establishes repeatable extraction of executable alternatives, with no replacement samples or execution-dependent repair. It does not establish an advantage over authored extension, whose qualification yield is identical. Incremental coverage on independently fixed tasks remains a separate test.

5.3 Measured data extension and strong baselines

Adding the complementary teaching encounter lets the learned policy solve a sixth development context that its five-context predecessor fails. A script reselected on the same six-context data also passes all six; the original script and strong constant each pass five (Table 5). Both updated policies choose WAIT at 0.30 and 1.00 s, then sustained adaptation at 1.40 s for the complementary passage, completing it in 4.54 s at seed 8732. This is an actual closed-loop result; the 4.20 s forced-teacher time at seed 8731 is a separate execution and is not substituted for the policy cost.

The comparison combines 20 original assignments on five contexts with 16 registered additions: the original four policies on the complement, and the updated learner and script on all six contexts. Only the learner’s teaching set changes; its phase-specific architecture and penalty 10 remain fixed. The script repeats the same 72-setting search on the enlarged development set. All 36 policy–context outcomes are known. Thirty-five have complete measurements; the complementary walking episode records an independently verified 385.218 N beam contact before a clip-wrap abort after 756 physics steps. That known failure remains in the denominator, with no later supervision or full-horizon imputation. The combined record contains 42,476 measured physics steps. Archived-source replay and a separate scorer compatibility audit preserve the original results.

The added passage carries a measurable cost tradeoff. Both updated policies choose sustained adaptation on the short and long passages, costing 0.50 s more in each than their respective predecessors: a mean increase of 0.20 s over the five mutually successful contexts. The updated learner and script have identical obstacle times and differ by one 0.02 s tick in the empty scene. Thus the learner uses the newly exposed complementary alternative but retains an avoidable cost regression and matches the strong script’s passage count. This comparison isolates neither replay nor construction. All layouts, including the complementary search and script selection, were used for development; a changed physics seed does not make them held out.

5.4 Can the learner retain capability without the cost regression?

Information-consistent continuation test. We first isolate a limitation that no learner can overcome without timely information. Mask all 28 corridor-summary features before a declared reveal decision, retaining recorded proprioception and legal actions. The remaining features are exactly equal across the six contexts at all three phases. With scene information available from 0.30 s or 1.00 s, an exact finite observation-group policy can pass all six contexts. Revealing it only at 1.40 s, or never, reduces the optimum to five, although the scene-wise schedule bank still covers six. At a 1.00 s reveal, the shared initial action is WAIT. At a 1.40 s reveal, the early context has lost its last prior commitment opportunity; no earlier common commitment also passes the complementary context. This recorded-table information ablation tests the teacher’s shared-action constraint; it does not simulate sensor latency or a new closed-loop rollout.

To separate the information value of WAIT from repertoire size, a matched control chooses once at 0.30 s among all seven complete schedules, including late-entry schedules. It executes neutral until the chosen entry. Within each initial information group, the control selects the schedule maximizing measured passage count, then minimizing summed successful passage time. Sequential selection can additionally condition later choices on newly delivered information. With corridor features revealed at 1.00 s, their finite passage limits are 5/6 and 6/6, respectively. With information present initially both reach 6/6; with reveal at 1.40 s or never both reach 5/6. These exact recorded-table limits retain the same action repertoire and all six assigned contexts. They identify when later information can help; they do not measure a learned or newly executed policy, and nominal singleton groups yield no demonstrated WAIT advantage.

Training the same ridge learner with group-consistent rather than scene-wise targets yields identical nominal coefficients. Both teachers produce 6/6 passing branch proxies with information by 1.00 s and 5/6 with later or absent information. In the latter conditions, group-consistent targets select the earlier prior, reducing mean recorded time by 0.004 s over the same five successes, but not recovering the complementary passage. Thus this development comparison establishes trainable information-consistent targets, not a passage advantage.

The alternative depth-two feasibility/time trees reduced fitted regret but passed only 4/6 actual development contexts, versus ridge’s 6/6, and were 0.12 s slower on their four mutually successful contexts. We therefore retain the common ridge learner. Appendix B.1 records the fixed-data control, native failures and observed input sensitivity.

Physical sensitivity of commitment and cost. A separate four-condition check executes the six-context policy on the short passage with nominal sensing, 30% channel dropout, 3 cm Gaussian range-noise standard deviation, and 100 ms packet latency. Each condition passes with zero measured beam force and legal recovery. Nominal sensing, dropout and range noise retain the 1.40 s sustained commitment and 4.10 s passage; latency instead selects the prior at 1.00 s and passes in 3.60 s. These one-context development results expose a timing-sensitive cost decision, not a rationale for delaying observations. Appendix C defines the interventions and their recorded-input comparison.

The information-group teacher finds six distinct exact student vectors at every phase, so it encounters no shared information groups in these six contexts. Controlled collision tests confirm that it rejects WAIT when indistinguishable future histories require incompatible actions, and retains WAIT when a later observation distinguishes them. The exact-data calculation does not establish tolerance to sensor noise. On the same whole-context folds, uniform, coverage, uniform/coverage and refreshed-error replay all retain four passing proxies out of six after three refits. More varied acquired encounters are needed to test the value of these learning mechanisms.

5.5 Acquisition breadth and reserved evaluation

The acquisition study compares feasibility-screened uniform, target-only, executed contrast, and the same executed scenes with observation-aware replay. Each arm has three independently acquired corpora. An empty bootstrap uses seven physical branches; every subsequent encounter adds one pre-update student and seven teachers. Analytic and observation arms share candidate identities and ordering within each seed, isolating replay from construction. Common-support quotas keep task-stratum counts matched, and known physical failures remain training data. Bootstrap recordings remain within their original corpus.

To test whether effects persist beyond individual teaching examples, the expanded study adds a reference-envelope contrast arm and the M8/M16/M32 checkpoints in Table 6. The completed original four-arm M0–M4 trajectories are retained; the reference arm acquires its own complete prefix. At M32, each of the 15 corpora is assigned seven short, seven sustained, seven early-constraint, six short–short and five short–sustained encounters.

Checkpoint	Completed encounters after bootstrap	Assigned episodes	Maximum steps
M2	2	23	27,416
M4	4	39	46,488
M8	8	71	84,632
M16	16	135	160,920
M32	32	263	313,496

Table 6: Learning-curve checkpoints per arm/corpus. Actual measured steps provide the learning-curve horizontal axis; assigned maxima are disclosed separately. Checkpoints share an acquisition trajectory rather than requiring independent reacquisition.

Acquisition is ongoing. The first broader closed-loop checkpoint will compare all 15 M8 policies with the strong script and all seven fixed schedules on the same six development contexts and physics seed: 138 assigned episodes. These measurements will separate selection failures from repertoire limitations on matched conditions. The fixed development continuation adds 90 policy episodes at M16 and 90 at M32, sharing the original 48 script/fixed captures: 318 unique evaluation episodes across the three checkpoints. Both later evaluations require completed M32 acquisition, so their outcomes cannot affect its stopping or scene queues.

The reserved benchmark contains 18 world-fixed layouts: 12 single constraints and six two-beam compositions. Each layout has two execution seeds and eight stress offsets. Layouts were fixed before option tuning, without clearance or outcome filtering. The proposed nominal comparison includes all 45 learned checkpoints and the script plus seven fixed schedules on 36 conditions each: 1,908 assigned episodes. Its implementation and assignment preview are complete; model freezing, evidence-based protocol adoption and execution remain pending. The eight original stress offsets remain reserved and disabled in this nominal comparison. Final policies and strong baselines will be frozen after development evaluation and before reserved execution. For each matched scene, seed and perturbation q , we report bank capability $C(q) = \max_s Y_s(q)$ separately from policy passage $P_\pi(q) = Y_\pi(q)$. All assigned tasks remain in the denominator, including tasks with no measured passing schedule. Unknown outcomes remain explicit; time differences use mutually successful matched episodes and report their counts. Baselines are shared comparisons, not additional independently trained corpora. The three training seeds provide corpus-level variation; they do not multiply the number of independent layouts. Reserved results are not yet available.

The earlier low-budget design and its paired analysis are retained in Appendix F, alongside acquisition implementation details.

6 Limitations and Conclusion

Motion2Scene connects useful motion alternatives to training scenes through executed geometry, complete physical continuations and causal sensor evidence. The measured six-context bank contains complementary successful responses. Adding the complementary teaching data raises the learner from five to six successful development contexts, matching the retuned script while adding 0.50 s on each of two earlier passages. The common-pool construction experiment also finds substantial reference-to-execution geometric disagreement. The completed M4 acquisition shows higher adaptation-required task yield for executed contrast, while each corpus remains solvable by a fixed schedule and its recorded branch-selection readout ties uniform. The central remaining test is whether execution-conditioned construction and causally usable supervision improve held-out traversal at matched acquisition cost; replay’s additional contribution is tested separately.

The current interface follows one finite route with at most one adaptation and recovery. It cannot stop, steer, select a route or chain repeated adaptations. Ideal ray normals, summarized rather than fully observed free space, reused source ancestry, nominal physical tests and a short fixed horizon limit the claims. Unknown torque prevents mechanical-work or battery-energy comparisons.

A Acquired Response Witness

The interim pool retains every audited M2 acquired encounter and adds all three assigned round-three executed-contrast encounters. Repeated geometry/seed pairs are merged only after their complete outcome vectors agree. This yields 20 conditions and one pair with disjoint passing sets. Both scenes use execution seed 93201: candidate 0155 was acquired by target-only at round one; candidate 0017 was acquired by executed contrast at round three. The pair was identified after inspecting outcomes and is an illustration, not a selected test set. The existing scene-wise teacher and 114-dimensional observation interface are unchanged. Exact recorded perception vectors differ at all three neutral-prefix decision phases; no observation tolerance or new memory rule is introduced. For example, the first-phase upper-hit band is 1.5–2.5 m in scene A and 0.75–1.5 m in scene B. These observations do not establish robust information separation under unmeasured noise or perturbations.

B Teacher Initialization and Decision-Learning Details

If a phase has no consequential row, initialization requires a complete matched table with at least two legal actions, all required continuations admitted, and all legal actions passing with exactly equal finite nonnegative passage times. Zero ridge coefficients and intercepts then solve the measured zero-regret targets. Normalization uses those rows with standard-deviation floor 0.05; only supported legal columns are trained. The deterministic lowest-index argmax can choose WAIT. Once consequential data exist, they alone determine the fit. Missing or all-failed tables do not initialize a head. This common rule was added after the first empty bootstrap exposed an exact cost tie at 1.00 s; the previously selected ridge penalty 10 was retained. Appendix F records the corresponding acquisition versions.

The September 9 decision-learning comparison uses all 42 existing development branches and leaves all phases of one context together in each fold. It compares the unchanged ridge with separate depth-two feasibility and time trees, both with deterministic seed zero. Feasibility uses measured failures as well as passes; time uses passes only. The loss and model family change jointly, so this comparison does not isolate nonlinearity. Exact feature keys canonicalize signed zero and otherwise preserve every stored float64 value. No measurement tolerance is imposed. All six keys differ at all three phases. Information partitions must refine over phases; full-state prefix hashes and scene identifiers are excluded from the observation-equivalence rule.

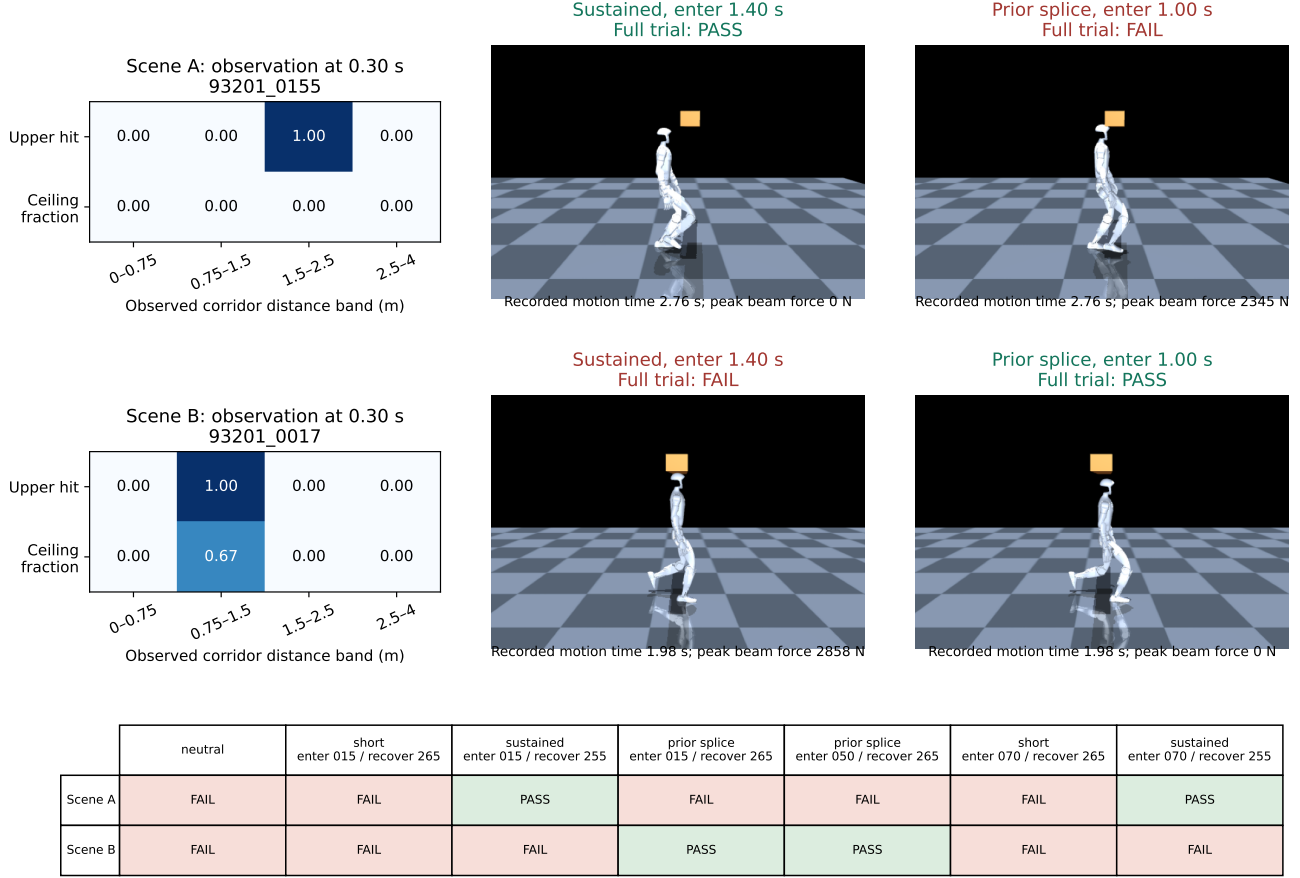
The replay comparison fits the same ridge on the same training portion of each fold. Each arm starts from a uniform fit and receives three refits. Current-error priority is recomputed exclusively on that fold’s training rows. Coverage uses the joint-bin definition in the main text. Held-out decisions never enter normalization, priorities or fitting. All four arms have four passing branch proxies out of six. Their mean consequential phase regrets are 0.18952 for uniform, coverage-only and uniform/coverage, and 0.19570 for current error. These controls balance encounters before phases, whereas the original ridge comparison weights available contexts equally within each phase; the two uniform conventions therefore need not have identical phase regret.

Neither CPU experiment adds physical execution. The existing teacher and model artifacts remain unchanged. Full per-fold choices, costs and replay weights are stored with the scripts `motion2scene_decision_study.py` and `motion2scene_replay_comparison.py`; the project development-learning report links the exact result files.

The completed M2 acquisition controls extend this comparison to three acquired corpora. Uniform, coverage-only, the actual 0.8-uniform/0.2-coverage fallback, historical gated gaps, historical ungated gaps, refreshed supervised error and historical-plus-error weights each select passing recorded branches on five of the same six development contexts in every corpus. Every corpus retains three positive historical phase gaps from its first acquisition student. All six nonempty phase cue records per corpus satisfy the current visibility rule, and gated and ungated replay weights are identical. Thus replay priority is active, but this checkpoint does not exercise a visibility exclusion or establish a passage gain. The historical gaps remain attached to their pre-update generating models. The refreshed residuals measure supervised fitting error. The 21 final control models require 66 complete policy fits: one common initial fit and three refits for each of seven controls, per corpus. These are CPU fitting operations, with no new physical evaluation.

The same seven controls are now complete at M4 for all three original corpora. All again select passing recorded branches in 5/6 contexts. Within each corpus, every control selects the same complete prior schedule on all six contexts: entry at 1.00 s for seeds 93201 and 93203, and 0.30 s for seed 93202. Relative to uniform, selected-schedule changes and passing-branch differences are zero; recorded time differences are also zero on the five mutually successful contexts per comparison. Each corpus retains three positive historical phase gaps, and all twelve nonempty phase cues satisfy the current visibility rule. Gated and ungated weights remain identical

An acquired pair requires different complete schedules



Complete measured bank outcomes; entry/recovery labels above use control ticks (50 Hz).

Recorded IsaacLab states rendered with MuJoCo forward kinematics only. Visual meshes differ from native collision bodies.

Figure 4: A complementary pair from the acquired development pool. Left: two recorded feature families at the first decision; the exported data retain all 114 features. Middle/right: recorded IsaacLab poses for two illustrative complete schedules, rendered using MuJoCo forward kinematics without dynamics integration. Each row uses the same control index, chosen at the failed displayed schedule’s maximum measured beam force. Rendered meshes differ from native collision bodies; the full-trial pass/fail labels and maximum contact forces come from the archived physical measurements. The complete seven-schedule outcomes are shown below. Contact failures here satisfy the unchanged environment-contact scoring rule. This post-hoc witness neither compares balanced acquisition arms nor measures a newly executed perceptive policy.

despite differences between replay and uniform weights. Re-evaluation of the 42 saved M2/M4 controls reproduces all 252 recorded context choices and 126 weight recipes without new fitting or physics. The checkpoints share nested training prefixes and the same six development contexts, so these are repeated readouts rather than independent policy trials. This extends the recorded-input replay null result; its separate closed-loop benefit remains unestablished.

B.1 Secondary learner comparison

The regression already appears on recorded teaching inputs: the six-context ridge selects sustained schedules on the short and long passages, with branch times 4.06 and 4.64 s. We compare it with depth-two decision trees predicting feasibility and successful passage time separately. Each legal action has a feasibility classifier; its time regressor uses successful observations only. Among actions with estimated passage probability at least 0.5, choose the fastest; if none qualifies, choose maximum estimated feasibility. Depth and threshold are fixed before this comparison, with no parameter search.

Both models select passing recorded branches in all six training contexts. The tree reduces mean regret over the 17 consequential phase tables from 0.01916 to 0.00713 and recovers the 4.48 s prior schedule on the long passage, but selects a slower 4.20 s schedule on the two-beam context. Leaving each entire context out gives both learners four passing branch proxies out of six; both miss the early and complementary contexts when those teaching examples are withheld. Thus the alternative improves some fitted decisions without establishing a generalization advantage. These are finite recorded-branch readouts, with no new closed-loop executions in the fitting comparison.

We then execute the same fixed tree on all six contexts at the policy-comparison seed 8732. Its exported decisions reproduce the original tree exactly, and its first neutral inputs match the ridge recordings in every context. The tree passes four contexts, losing the long and complementary passages to measured contact; ridge passes all six. Tree passage times are 5.90, 4.06, 2.72 and 4.12 s on the empty, short, early and two-beam contexts, respectively. Over these four mutually successful contexts it is 0.12 s slower on average. All six measurements are complete. On the complementary context, a small change in the distant corridor’s unknown fraction crosses a feasibility-tree threshold and causes immediate short commitment instead of WAIT. Thus lower fitted regret does not resolve the closed-loop cost/capability tradeoff. We retain ridge as the common acquisition learner and expand the teaching corpus rather than selecting this alternative.

C Sensor Sensitivity Implementation and Development Check

At each capture, dropout removes fan channels independently without adding free-space evidence for missing measurements. Gaussian noise perturbs detected surface ranges; a distance outside the sensor interval becomes a missing return. Surviving normals and robot pose remain ideal. Fixed latency delays the corrupted packet before its world-space rays update the observation history. Capture and delivery times remain separate; the student receives neither clean rays nor future packets. Random draws are indexed by corruption seed, frame and fixed fan-channel position, independently of which channels hit a surface.

On recorded neutral prefixes from all six teaching contexts, zero perturbation reconstructs all 18 decision vectors exactly. At corruption seed 95001, each of 10%/30% dropout, 1/3 cm range noise and 40/100 ms delay changes the short-context branch proxy from sustained to prior, while preserving six passing proxies. The separate native four-condition check uses the existing short scene, the six-context policy, physics seed 8732 and corruption seed 95001. It records four complete episodes, 4,768 physics steps, and 1,192 exactly reconstructed student feature vectors. All conditions pass with zero measured beam force. Nominal, 30% dropout and 3 cm noise retain sustained and take 4.10 s; 100 ms latency selects prior and takes 3.60 s. Thus the dropout/noise proxy change from the seed-8731 teaching capture does not reproduce in this seed-8732 execution. This controlled development check exercises the perception interventions; multi-layout, multi-seed sensitivity of the selected final policy remains part of the broader evaluation.

D Historical Development and Design Decisions

The original diagnostic manuscript, `paper.tex`, and every earlier working draft remain unchanged or archived. Their results explain the present interventions but do not supply an untouched evaluation of them.

The original command ceiling. The earlier 36-condition walk/d040 bank contains 14 both-pass, ten crouch-only, 12 both-fail and no walk-only conditions. Its maximum passage under those two fixed commands is $24/36 = 66.7\%$, already attained by always-d040 and the analytic selector. A learned selector misses two otherwise solvable conditions. This separates missed decisions from missing capability: replacing a selector cannot solve a condition for which every unchanged command fails. The bank was extensively inspected during development and remains a regression benchmark.

Execution and observation interventions. Four-second development experiments qualified additional authored crouch references and legal entry times. A newly constructed beam defeated walk and d040 while d055, d070 and d085 passed. The old low beam still defeated every tested option. These references are `local_crouch` edits of a generated neutral source, not independently learned skills. Their old fixed return windows do not qualify the present six-second durations.

Identical earlier 214-dimensional inputs sometimes have conflicting walking labels. A larger deterministic classifier cannot distinguish identical inputs. All members of the observed ambiguous classes remained passable with the conservative adaptation; the ambiguity concerned unnecessary adaptation rather than unavoidable passage failure. The later floor/ceiling history preserves information that a single surface-height value discards, but does not guarantee unique task labels, semantic obstacle identity or timely underside observations.

A six-second 106D duration experiment executes three options on three scenes: empty permits all, a short beam permits short/sustained, and a long passage permits sustained only. Its separate 15-policy evaluation at seed 8732 has learned and scripted policies both passing three of three with identical times 5.40/4.10/4.68 s; both select sustained on the short beam although constant short passes in 4.06 s. Constant sustained fails the empty finite-horizon hold without contact or fall. These outcomes establish a duration distinction and an integration limitation; changes from 106D/0.5 s memory to 114D/2 s memory are not a controlled sensor ablation. Earlier four-second phase-head data updates yield a one-control-tick cost change on one development condition, while a shared-head update regresses aggregate cost. Neither is a primary curriculum result.

Learned proposals remain a negative development result. An earlier learned conditional proposal uses paired executed summaries and a 15-frame approach history. In its final geometric pilot it accepts 41 of 192 proposals against 123 of 192 for analytic resampling and zero for uniform sampling. This pilot neither demonstrates neural acquisition advantage nor measures the new curriculum’s held-out traversal. The present primary method uses an analytic constructor. A future conditional proposal could approximate its useful regions; a future adversary could edit geometry while retaining a physically verified solution. Neither extension is executed in the primary fixed candidate queues.

Prior sample and authored derivative. The installed Kimodo constrained experiment generates one six-second sample using the existing checkpoint. It violates native joint limits and proposed joint checks; the raw sample is retained as rejected. An explicitly registered operator projects 11 violating joint samples and splices the qualified neutral prefix and tail with smooth transitions, without reinference or hidden clipping. Native kinematic flags, including foot-ground geometry and loader interpolation at the splice boundary, remain recorded. The derived reference is admitted only after complete physical schedule qualification. Its successful early-beam response cannot be attributed to unmodified neural generation alone. A separate CPU SONIC planner sample establishes only installed inference and conversion; it is not a physically qualified member of the current option bank.

Contact and scorer revisions. The original strict net-contact predicate rejected the six-second motions because of inherited hip-wrist contacts, including in neutral walking. A separately registered counterpart experiment measures self-contact and environmental contact independently; new physical captures qualify under the stated external-contact criterion. Old strict failures remain failures under their original predicate. No criterion revision is presented as an unchanged safety contract.

The single-beam scorer initially omitted the declared initial-upstream condition already checked by courses. The explicit repair requires every initial body origin to lie upstream of the beam entry plane. With complete measurement/source admission, its violation is named `invalid_initial_approach`, a task failure distinct from collision or fall; missing evidence remains unknown. The applied repair passes 70 focused tests. Independent replay of 45 prior recordings changes no passage, time or classifier outcome; their minimum initial upstream

margin is 0.842 m. Subsequent common learner/runtime freezes bind the new scorer, while original execution snapshots remain preserved.

E Portable Dataset and Reproducibility Details

The data separate deployable input arrays from privileged supervision and provenance. Each episode contains pickle-free state trajectories, raw measurement-only rays, named features and per-packet alignment masks; pre-action/commitment/return records; all-body 200 Hz environment and beam contact streams with counterpart ordering; and separate pass/failure/unknown costs. Recorded switches determine the executed option, rather than the configured preferred option. Complete WAIT targets map to continuation episode IDs.

The 42-episode 114D snapshot supports reconstruction of 18 phase tables and fitting with bundled data and NumPy alone. At penalty 10, it reproduces all 18 teacher-table decisions and 84 archived neutral policy actions; the largest coefficient difference is 4.4×10^{-17} . Standalone validation also passes with private motion assets, production sources and simulator imports blocked. A separate portable panel contains all 36 actual policy-comparison episodes, including the known partial failure: 10,619 physical rows, 10,618 aligned 114D packets and 42,476 physics steps. Its two source-version bundles retain original and corrected scorer provenance. These policy recordings contain no synthetic teacher targets and complement the forced-branch teaching release.

The completed original four-arm M2 acquisition release retains twelve separate corpora, 252 teacher branches and 24 actual pre-update student executions: 276 episodes, 328,992 measured physics steps and 82,248 aligned sensor rows. All 52 teacher and nine student failures remain included. Its NumPy reader reconstructs 108 complete phase tables and all 36 frozen M0/M1/M2 ridge fits; all 324 recorded teacher-packet action checks agree, with maximum coefficient difference 5.6×10^{-17} . These checks reuse recorded packets and do not measure newly executed policy passage. The reader also reproduces the M1/M2 physical task-yield and acquisition-cost tables from the portable episode manifests. Replay reconstruction uses the original audited historical weights, not a new physical-gap measurement. Both the original export and a relocated archive extraction pass with original experiment reads and simulator imports blocked by a Python-level validation guard. The original startup assessment, zero measured startup steps, separate 1,192-step conservative reservation and failed bootstrap fit remain preserved; the seven reused physical bootstrap captures stay in their original corpus and are counted once.

Later five-arm export/reconstruction utilities preserve inherited fit prefixes and the expanded fitter’s distinct historical-replay sidecar format. Their NumPy reader reproduces the complete M2 release and three native expanded-format fits on recorded M2 data, with 27/27 matching recorded action checks for the latter. These are CPU artifact checks, not M8 acquisitions or new policy executions. Later-checkpoint exports remain gated on their completed physical prefixes; unsupported partial records are rejected explicitly rather than discarded from an assigned corpus.

The primary reuse target is the 114D seven-schedule data, with separate physical labels, exact neutral history and complete future continuations. The six-context development package preserves all 42 episodes, including 14 physical failures, 12,516 aligned packets, 50,064 physics steps and 18 phase tables. Seventeen tables provide consequential successful targets. Original criterion/source identities travel with each episode; later upstream-scorer compatibility receipts are separate evidence, not substituted execution code. Independent release audits and standalone fitting receipts are linked in the project documentation.

The source collections and derived package are recorded under these artifact IDs:

- `m2s-timed-schedule-five-context-teachers-v1`;
- `m2s-complementary-schedule-teachers-v1`;
- `m2s-timed-schedule-six-context-dataset-v1`.

The actual five-context policy validation and independent replay are:

- `m2s-seven-schedule-policy-comparison-v1`;
- `m2s-seven-schedule-policy-comparison-audit-v1`.

The manuscript’s evidence receipt pins these paths and hashes. These are portable workspace releases, not a claim that a public hosting or redistribution license has been finalized. Upstream asset terms are retained, and pretrained motion or controller weights are excluded.

Each episode records its source and option ancestry, registered and actual command, physical seed, native geometry and contact counterpart mapping, original capture identity and hash, and exact postdecision switches separately from student inputs. Original capture path plus hash identifies an acquisition: byte-identical independently executed trajectories are distinct episodes. Conversely, symlinked analysis views of one capture never create additional executions. Unknown startup physics counts remain unknown; a finite unmeasured reservation is not converted into a measured count.

Earlier schemas remain separately versioned. The corrected legacy 214D package contains 72 encounter groups and 144 branches, including the original regression and nominal acquisition pairs; it is development data. Four-second 100D and 110D option packages, the 106D duration recordings, and the current 114D schedules are not concatenated as one homogeneous learner input. Aligned releases retain all raw packets plus eligibility masks. In the affected 199-row recordings, the last sensor packet follows a command reset and cannot be aligned to the last valid physical row; it remains present but masked. Complete modern bounded six-second captures avoid that wrap and retain 298 exact aligned packets; partial failures retain their actual shorter physical and sensor sequences. Old archives are immutable and their errata identify corrected releases.

The dataset supports reproducible finite-schedule value learning, feasibility prediction, causal sensing studies and downstream option selection in overhead traversal. It contains no steering or route-choice demonstrations. Calling it a general indoor navigation dataset would exceed its action and scene coverage.

Acquisition ledger V8 adds the original seven primary empty-bootstrap captures once to V7’s development inventory. Its fixed snapshot contains 232 recorded study attempts: 231 complete episodes with 233,624 physics steps and one known partial failure with 756 steps, totaling 234,380 study steps. The separate retained instrumentation failure adds 796 steps, giving 235,176 persisted steps in the declared scope. Four separately inferred unpersisted steps and unknown startup counts remain separate. The earlier pre-simulator attempt retains its 1,192-step reservation and zero recorded physics, outside these capture totals. The seven added captures contribute 8,344 steps; archived-code reuse, fitting and exporting add none. Historical 144-branch data and primary captures acquired after this fixed snapshot are excluded. This is bookkeeping across research stages, not a matched-budget constructor comparison. The current 36-row policy panel reuses its original 20 assignments and does not count them a second time.

F Primary Protocol, Provenance and Remaining Evidence

The earlier low-budget evaluation proposal pairs 24 M2/M4 models and three shared baselines with 18 nominal layouts and seeds 94301/94302: 972 episodes and at most 1,158,624 evaluation physics steps. Baselines are walking, one development-selected constant and one development-selected multi-option sensor script. Missing acquired models cannot be replaced with development models. The additional seven-schedule capability panel and eight stress offsets per layout are not included in that 972-episode count. The expanded five-arm study now precedes final evaluation selection; the earlier design remains preserved as a proposal, not a completed benchmark result.

The low-budget analysis specifies 20,000 paired crossed corpus/layout bootstrap draws, keeping each layout’s two execution seeds together. Baselines are shared once, not replicated as independent corpus trials. Per-corpus means and variation accompany descriptive intervals; there are only three corpus replicates. Unknown outcomes withhold intervals. Time comparisons use mutually successful matched episodes and state their counts. Neither a different physics seed nor a separate model fit makes an inspected development layout held out.

The immutable acquisition plan V3 preselects a common supported set of four strata for each of three proposal/physics seeds 93201–93203. Uniform, target-only and analytic queues come from the same newly drawn candidate pools; coverage shortages and omitted strata remain explicit. The observation arm shares the analytic queue exactly. The comparison therefore concerns construction conditional on common stratum support; it does not estimate unconstrained coverage over every requested scene family.

The controller starts from seven independently acquired bootstrap schedules within each corpus, then records one pre-update student and seven teachers per round. The adopted continuation retains the original analytic seed-93203 bootstrap in that same corpus only, with no second acquisition charge or cross-corpus reuse. Models M0–M4 use only the allowed preceding/current teacher collections. The observation arm performs four additional CPU-only unweighted audit fits before its weighted fits; these add compute but no branches. Historical replay binds every student’s generating model, its prior training collections, exact matched histories and age. The pre-adoption replay amendment explicitly corrects earlier “uniform fallback” prose to the implemented 0.8-uniform/0.2-coverage mixture when no positive gap exists. The positive-gap mixture remains 0.2/0.2/0.6. Fixed source files and scene

queues are unchanged by that disclosed amendment.

Every new physical launch requires exact runtime, native asset, environment, model and scene preflight. A nonblocking shared lock and immutable launch intents preserve serial acquisition. Known physical failures remain assigned outcomes; unresolved intents or unknown measurements pause without automatic retry and reserve the remaining finite horizon. Dispatch ordering interleaves all 12 corpora at each acquisition boundary. The reviewed dispatcher V2 required a separate source/adoption-binding and parent-lock repair. Its versioned V3 repair passes 11 independent mocked dispatch tests; this is software evidence and creates no execution authority. The original consequential-only fitter stopped at its first primary bootstrap because the 1.00 s legal actions had exactly equal successful costs. Both the failed fit and its teacher table are retained.

The separately adopted common learner amendment enables measured-tie initialization only at a phase lacking consequential support, under the complete all-success conditions in the main method. The first bootstrap and CPU validation of that same table informed this change; no primary contrast encounter, primary student rollout or reserved outcome informed it. The comparison is therefore prospective under a disclosed method amendment, not a fresh confirmatory test of an untouched learner. All-failed, missing or unmatched tables supply no targets; fitting stops only if a phase has neither consequential support nor qualifying tied support. The applied implementation passes 179 focused tests (106 impacted checks and 73 tie/provenance checks). Source freeze V4 and native asset/environment freeze V6 bind the adopted code and installed assets; these checks provide implementation evidence, not robot performance.

The V4 proposed reserved evaluation adds M2 to the original M4 comparison without new acquisition. Its original proposal hash is 966d45ed93066a73a945a55f33c769459fe4c1736a455d4eec4ba581d40de2fd; geometry remains the original fixed-world six-second V3 panel. A separate reporting clarification declares paired crossed corpus/layout resampling as the primary descriptive interval, retaining layout-only resampling as sensitivity. This explicitly amends the original layout-only uncertainty prose before primary acquisition or reserved outcomes.

All 34 paired comparisons are fixed: observation minus analytic, analytic minus uniform, and analytic minus target-only at each checkpoint; every arm/checkpoint minus each of three shared baselines; and M4 minus M2 within each arm. One RNG produces corpus-index draws before layout-index draws and reuses their weights throughout. Both evaluation seeds stay in each layout block. Time resampling uses the weighted matched-success numerator/denominator. If any draw has no jointly successful support, that time interval is withheld instead of discarding the draw. Unknown assigned outcomes withhold both completion and time intervals. No multiple-comparison significance claim is made.

Reserved execution requires a separate adoption record with evidence for all nine gates: complete schedule qualification; exact 114D runtime smoke; two-beam full-horizon execution; tested terminal/failure scoring; integrated protocol validator; actually validated strong multi-option script; every model/threshold frozen; complete source/assets/environment freeze; and no prior reserved outcome inspection. The model gate includes all 24 exact M2/M4 slots from 12 independently acquired corpora and all three baseline definitions. The complete audit of the 16-episode six-context development extension is complete, including one known partial walking failure. Adoption V3 binds that validation, the final script selection, refreshed common/native freezes, dispatcher V5, the replay amendment and measured-tie initialization. All 12 configuration checks pass. The earlier scene-resolution failure happened before Python/Isaac and retains zero recorded physics plus its original unknown assessment and 1,192-step reservation. The subsequent original empty bootstrap contains five passes and two sustained-option finite-horizon hold failures. Its original M0 fit failed without changing those physical outcomes.

A dedicated Python-isolated worker imports the original archived collector, executes its exact nonphysical resolver checks and independently recomputes all seven original rows without writing their artifacts. The original-interpreter receipt verifies 8,344 recorded steps. The new controller binds this audit and reuses those captures only in their original corpus; later refits do not create new acquisitions. At 04:34:26 UTC on 9 September 2026, dispatcher boundary zero completed the first primary M0 from these data, with consequential targets at 0.30/1.40 s and one measured-tie initializer at 1.00 s. The model SHA-256 begins 925a4e2e890c2a07; the full hash is pinned in the project status receipt. This is an actual controller-produced primary model, distinct from the earlier external CPU validation model. No M2/M4 traversal or independent curriculum comparison is reported at this first-model status cutoff.

The current reserved V6 proposal and reporting V3 update exact source/model provenance while retaining the 972 V4 assignment identities, order and clarified statistics. They remain unadopted and unexecuted. Known measured physical failures can complete an assigned slot even when later supervision is unavailable. No unknown outcome is filled from a geometric screen, teacher proxy, package fit or an unverified partial execution.

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